Information Fusion 24 (2015) 72–83

Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus

The fusion process with heterogeneous preference structures in group decision making: A survey

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ARTICLE INFO

Article history:
Received 3 August 2014
Received in revised form 15 November 2014
Accepted 16 November 2014
Available online 25 November 2014

Keywords:
Group decision making
Heterogeneous preference structures
Fusion process
Consensus

ABSTRACT

In group decision making (GDM), decision makers who have different experiential, cultural and educational backgrounds will naturally provide their preference information by heterogeneous preference structures (e.g., utility values, preference orderings, numerical preference relations and multigranular linguistic preference relations). To date, many studies have discussed GDM problems with heterogeneous preference structures. To provide a clear perspective on the fusion process with heterogeneous preference structures in GDM, this paper presents a review of three types of fusion approaches: the indirect approach, the optimization-based approach and the direct approach. Moreover, with respect to insights gained from prior researches, several open problems are proposed for the future research.

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1. Introduction

Group decision making (GDM) consists of the interaction of multiple individuals to rank a set of alternatives from best to worst (or select the most desirable alternative(s)). To obtain the final solution, two processes are often needed [1,2]: (1) the consensus process; and (2) the selection process. The consensus process refers to obtaining the maximum consensus degree among decision makers on the solution set of alternatives. The selection process aims to aggregate individuals' preference opinion into a collective opinion, and then rank alternatives for selecting the most desirable alternative(s) based on the obtained collective opinion.

Many approaches have been proposed to solve GDM problems (e.g., [3–8]). In these approaches, the preference information provided by decision makers is represented by the same preference structure. In practical GDM problems, however, it is quite natural for decision makers to have different experiential, cultural and educational backgrounds. As a result, different decision makers may use different preference structures to express their individual preference information. Commonly used preference structures are utility values [9,10], preference orderings [9–11], multiplicative preference relations [9,12,13], fuzzy preference relations [9,10,14], and linguistic preference relations [1,4,6,15–18]; thus far, many approaches have been developed to discuss the fusion process with heterogeneous preference structures in GDM, which can be classified into three main research lines:

(1) The indirect approach, in which heterogeneous preference structures used by decision makers are transformed into a uniformed preference structure. The indirect approach proposed in [9,10,12,17–23] consists of three steps: (1) unifying the heterogeneous preference information provided by decision makers through transformation functions; (2) aggregating individuals' preference information with the uniformed preference structure into a collective one; and (3) ranking alternatives for selecting the most desirable alternative(s).
(2) The optimization-based approach [24–27], in which different multi-objective optimization models are constructed to integrate preference information with heterogeneous preference structures.
(3) The direct approach, proposed by Herrera et al. in [28]. Afterwards, a direct consensus framework for GDM problems with heterogeneous preference structures was proposed by Dong and Zhang [29] to reach a high consensus degree among decision makers on the solution set of alternatives.

The three types of fusion approaches have different characteristics: (1) in the indirect approach, transformation functions are used to unify the heterogeneous preference structures; (2) in the optimization-based approach, multi-objective optimization models are key elements to integrate heterogeneous preference structures; and (3) in the direct approach, the collective preference vector is obtained by aggregating individual preference vectors. To provide a clear perspective on the fusion process with heterogeneous preference structures in GDM, this paper will present a review of these three types of fusion approaches, i.e., the indirect approach, the

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http://dx.doi.org/10.1016/j.inffus.2014.11.003
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optimization-based approach and the direct approach, in GDM with the heterogeneous preference structures. Moreover, with respect to insights gained from prior researches, we aim to propose new directions for future research.

The rest of this paper is organized as follows. We review the GDM problem with heterogeneous preference structures in Section 2. Next, we introduce the resolution framework of the indirect approach and the transformation functions among heterogeneous preference structures in Section 3. Following this, we survey the recently developed optimization-based approach in Section 4. Subsequently, we investigate the direct approach and its applications in Section 5. Additionally, we note several critical comments on the three types of fusion approaches and new directions for future research in Section 6. Finally, we draw a number of conclusions in Section 7.

2. The GDM problem with heterogeneous preference structures

In this section, we introduce the GDM problem with heterogeneous preference structures.

Let $X = \{x_1, x_2, \ldots, x_n\}$ ($n \geqslant 2$) be a finite set of alternatives, where x_i represents the ith alternative, and let $D = \{d_1, d_2, \ldots, d_m\}$ ($m \geqslant 2$) be a finite set of decision makers, where d_k represents the kth decision maker. For each decision maker, $d_k \in D$, has his/her own experiential, cultural and educational background, and it is quite natural to think that different decision makers will express their preference information using different preference structures.

In this situation, the decision problem is how to rank alternatives for selecting the most desirable alternative(s) among a finite set X based on the heterogeneous preference information provided by decision makers.

Decision makers' preference information over alternatives, X, is usually assumed to be represented by one of the following preference structures:

(1) **Utility values** [9,10]: In this preference structure, the decision maker d_k gives his preferences on X as a set of n utility values, $U^k = \{u_1^k, u_2^k, \ldots, u_n^k\}$, where $u_i^k \in [0, 1]$ represents the utility evaluation given by the decision maker d_k to the alternative x_i. The larger u_i^k value indicates the higher preference degree to the alternative x_i.

(2) **Preference orderings** [9–11]. In this preference structure, the decision maker d_k gives his preferences on X as an individual preference ordering, $O^k = \{o^k(1), \ldots, o^k(n)\}$, where $o^k(\cdot)$ is a permutation function over the set, $\{1, \ldots, n\}$. For example, suppose that a decision maker d_k provides his/her preferences on alternatives $X = \{x_1, x_2, x_3\}$ as the following preference ordering $O^k = \{3, 1, 2\}$. This means alternative x_2 is the best for d_k, while alternative x_1 is the worst.

Moreover, preference relations are the most widely used preference representation structures in GDM, which can be classified into two categories: numerical preference relations and linguistic preference relations.

(3) **Numerical preference relations**

There are two well-known types of numerical preference relations: Fuzzy preference relations and multiplicative preference relations.

(i) **Fuzzy preference relations** [9,10,14]. In this preference structure, the decision maker d_k gives his preferences on X as a fuzzy preference relation $P^k = \left(p_{ij}^k\right)_{n\times n}$, where $p_{ij}^k \in [0, 1]$ denotes the preference degree of the alternative x_i over x_j: $p_{ij}^k = \frac{1}{2}$ indicates indifference between x_i and x_j; $p_{ij}^k = 1$ indicates that x_i is absolutely preferred to x_j; $p_{ij}^k > \frac{1}{2}$ indicates that x_i is preferred to x_j. The fuzzy preference relations matrix is usually assumed to be additive reciprocal, but not necessarily. In this paper, we assume that the preference matrix, P^k, is additive reciprocal, i.e., $p_{ij}^k + p_{ji}^k = 1$ for $\forall i, j$.

(ii) **Multiplicative preference relations** [9,12,13]. In this preference structure, the decision maker d_k gives his preferences on X as a multiplicative preference relation $A^k = \left(a_{ij}^k\right)_{n\times n}$, where a_{ij}^k indicates a ratio of the preference degree of alternative x_i over x_j, i.e., it is interpreted as x_i is a_{ij}^k times as good as x_j. The preference matrix, A^k, is assumed to be multiplicative reciprocal, i.e., $a_{ij}^k \cdot a_{ji}^k = 1$ for $\forall i, j$.

Different numerical scales are used in multiplicative preference relations, such as the Saaty scale [13], the Ma-Zheng scale [30], the geometrical scale [31] and the Salo-Hämäläinen scale [32]. For example, Saaty [13] suggests measuring a_{ij}^k using a ratio scale 1 to 9: $a_{ij}^k = 1$ indicates indifference between x_i and x_j, $a_{ij}^k = 9$ indicates that x_i is absolutely preferred to x_j, and $a_{ij}^k \in \{2, 3, \ldots, 8\}$ indicates intermediate evaluations.

(4) **Linguistic preference relations**

The basic notations and operational laws of linguistic variables are introduced in [33–39]. Let $S = \{s_j | j = 0, \ldots, g\}$ be a linguistic term set with odd granularity $g + 1$, where the term s_j represents a possible value for a linguistic variable. It is usually required that the linguistic term set satisfies the following characteristics:

(i) The set is ordered: $s_i \leqslant s_j$ if and only if $i \leqslant j$.
(ii) A negation operator: $neg(s_j) = s_{g-j}$.

Generally, the semantics of the terms in S is given by type-1 fuzzy sets, which are described by a membership function, such as linear trapezoidal membership function [40], linear triangular membership function [41], Gaussian functions [42] and so on. In recent years, interval type-2 fuzzy sets have been proposed to define the semantics of terms in S [33]. Furthermore, different decision makers can use multiple linguistic term sets with distinct granularities to express their individual preference information.

Solving a decision problem using linguistic information implies the need for computing with words (CWW) [33–39]. Herrera and Martínez [43] initiated a notable symbolic model for CWW, called 2-tuple linguistic modeling.

Let $S = \{s_j | j = 0, \ldots, g\}$ be a linguistic term set with the granularity $g + 1$ and $\beta \in [0, g]$ be a value supporting the result of a symbolic aggregation operation; then, the 2-tuple that expresses the equivalent information to β is obtained by the following function: Δ: $[0, g] \to S \times [-0.5, 0.5)$, where

$$\Delta(\beta) = (s_j, \alpha), \text{ with} \begin{cases} s_j & j = \text{round}(\beta) \\ \alpha = \beta - j & \alpha \in [-0.5, 0.5) \end{cases}. \tag{1}$$

Clearly, Δ is the one to one mapping function. For convenience, its range is denoted as $\bar{S}$. Then, the inverse function of Δ is $\Delta^{-1}((s_j, \alpha)) = j + \alpha$. A 2-tuple negation operator presented in [37,43] is: $neg((s_j, \alpha)) = \Delta\,(g - (\Delta^{-1}(s_j, \alpha)))$.

In [44,45], linguistic preference relations based on a linguistic 2-tuple are defined as follows: the matrix $L = [l_{ij}]_{n\times n}$, where $l_{ij} \in S$, is called a simple linguistic preference relation, and the matrix $L = [l_{ij}]_{n\times n}$, where $l_{ij} \in \bar{S}$, is called a 2-tuple linguistic preference relation. If $l_{ij} = neg(l_{ij})$ for $\forall i, j$, then L is considered reciprocal.

In this preference structure, using the pre-defined linguistic term set S, the decision maker d_k gives his preferences on X as a linguistic preference relation $L^k = \left[l_{ij}^k\right]_{n\times n}$, where l_{ij}^k estimates the preference degree of the alternative x_i over x_j. Initially, the decision

maker makes pairwise comparisons using the terms of S, and constructs simple linguistic preference relations. The 2-tuple linguistic preference relations only appear in operations.

In the following sections, we will review three types of fusion approaches, i.e., the indirect approach, the optimization-based approach and the direct approach, respectively, to provide a clear perspective regarding the fusion process with heterogeneous preference structures in GDM.

3. The indirect approach

In this section, we review the indirect approach for GDM with heterogeneous preference structures. Specifically, Section 3.1 presents the resolution framework of the indirect approach, Section 3.2 discusses the transformation functions among the heterogeneous preference structures, and Section 3.3 introduces the transformation functions among the multigranular linguistic term sets.

3.1. The resolution framework of the indirect approach

As decision makers provide their preference information with heterogeneous preference structures, the first step of the resolution process in the indirect approach is to normalize the heterogeneous preference information. The selection process is then used to select the most desirable alternative(s) according to the uniformed preference information.

A general resolution framework of the indirect approach [9,10,12,17–23] is shown in Fig. 1. In this resolution framework of the indirect approach, the implementation of GDM with heterogeneous preference structures comprises a three-step procedure:

(1) Normalizing the preference information via the transformation functions. Transformation functions are essential to unify the heterogeneous preference information in the indirect approach. Additionally, many of the transformation functions among different preference structures have been proposed. In Section 3.2, we introduce the transformation functions among heterogeneous preference structures in further detail.
(2) Application of a consensus process. Naturally, at the beginning of every GDM problem, decision maker's opinions may differ substantially. Therefore, it is necessary to develop a consensus process in the indirect approach to obtain a consensus solution. The corresponding consensus models based on the indirect approach have been proposed in [10,46–48].
(3) Application of a selection process. Once the uniformed preference information with consensus is obtained, the selection process is carried out to rank alternatives for selecting the best alternative(s). The selection process in the indirect approach is developed in two phases:
   (i) Aggregation phase. In this phase, uniformed individual preference information is aggregated into the collective preference information by means of an aggregation operator. The most widely used aggregation approaches are fuzzy majority and ordered weighted averaging operator [9,49–51].
   (ii) Exploitation phase. In this phase, the rankings of alternatives (or the best alternative(s)) are obtained from the collective preference information, based on two choice degrees: the quantifier guided dominance degree and the quantifier guided non-dominance degree [52].

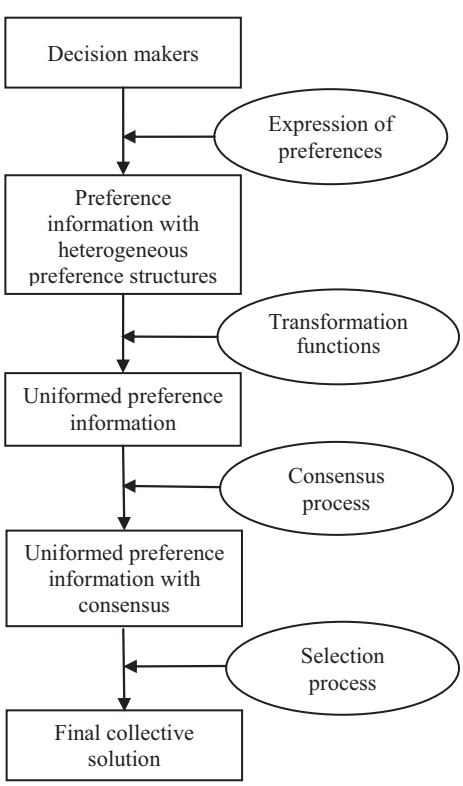

Fig. 1. The resolution framework of the indirect approach.

3.2. Transformation functions among heterogeneous preference structures

In this subsection, we introduce the different transformation functions among the following preference structures: utility values, preference orderings, fuzzy preference relations, different types of multiplicative relations, and linguistic preference relations.

(1) **Transformation functions between utility values and preference relations**

Utility values can be transformed into fuzzy preference relations [9] and multiplicative preference relations [21].

Let X and D be as before. Suppose that decision maker d_k provides his/her preferences on X by means of utility values $U^k = \{u_1^k, u_2^k, \ldots, u_n^k\}$.

(i) Transforming utility values into fuzzy preference relations. Chiclana et al. [9] proposed a general transformation function between utility values and fuzzy preference relations. For decision maker d_k, using transformation function TF_u^f to obtain his/her fuzzy preference value of the alternative x_i over x_j, p_{ij}^k, from utility values u_i^k and u_j^k, as Eq. (2):

$$p_{ij}^k = TF_u^f\left(u_i^k, u_j^k\right) = \begin{cases} \frac{\varphi_1\left(u_i^k\right)}{\varphi_1\left(u_i^k\right)+\varphi_1\left(u_j^k\right)} & \text{if } \left(u_i^k, u_j^k\right) \neq (0,0) \\ \frac{1}{2} & \text{if } \left(u_i^k, u_j^k\right) = (0,0) \end{cases}, \tag{2}$$

where φ_1: $[0,1] \to \Re^+$ can represent any non-decreasing and continuous function.

When setting $\varphi_1(u_i^k) = u_i^k$ [53–55], the transformation function TF_u^f can be represented as $TF_u^f(u_i^k, u_j^k) = \frac{u_i^k}{u_i^k+u_j^k}$. Similarly,

when setting $\varphi_1(u_i^k) = (u_i^k)^2$ [56], the transformation function TF_u^f can be represented as $TF_u^f\left(u_i^k, u_j^k\right) = \frac{(u_i^k)^2}{(u_i^k)^2 + \left(u_j^k\right)^2}$.

(ii) Transforming utility values into multiplicative preference relations.
Herrera et al. [21] proposed a general transformation function between utility values and multiplicative preference relations.
For decision maker d_k, using transformation function TF_u^m to obtain his/her multiplicative preference value of the alternative x_i over x_j, a_{ij}^k, from utility values u_i^k and u_j^k, as Eq. (3):

$$a_{ij}^k = TF_u^m\left(u_i^k, u_j^k\right) = \frac{\varphi_2(u_i^k)}{\varphi_2\left(u_j^k\right)}, \tag{3}$$

where φ_2: $[0,1] \to \Re^+$ can represent any non-decreasing and continuous function.
When setting $\varphi_2(u_i^k) = u_i^k$ [13,57], the transformation function TF_u^m can be represented as $TF_u^m\left(u_i^k, u_j^k\right) = \frac{u_i^k}{u_j^k}$. Similarly, when setting $\varphi_2(u_i^k) = (u_i^k)^2$, the transformation function TF_u^m can be represented as $TF_u^m\left(u_i^k, u_j^k\right) = \frac{(u_i^k)^2}{\left(u_j^k\right)^2}$.

(2) **Transformation functions between preference orderings and preference relations**

Preference orderings can be transformed into fuzzy preference relations [9] and multiplicative preference relations [21].

Let X and D be as before. Suppose that the decision maker d_k provides his/her preferences on X by means of preference orderings $O^k = \{o^k(1), \ldots, o^k(n)\}$.

(i) Transforming preference orderings into fuzzy preference relations.
Chiclana et al. [9] proposed a general transformation function between preference orderings and fuzzy preference relations.
For decision maker d_k, using transformation function TF_o^f to obtain his/her fuzzy preference value of the alternative x_i over x_j, p_{ij}^k, from values $o^k(i)$ and $o^k(j)$, as Eq. (4):

$$\begin{aligned} p_{ij}^k &= TF_o^f(o^k(i), o^k(j)) \\ &= \frac{1}{2}[1 + \varphi_3(o^k(j) - o^k(i)) - \varphi_3(o^k(i) - o^k(j))], \end{aligned} \tag{4}$$

where φ_3 is any non-decreasing function.
When setting $\varphi_3(o^k(i), o^k(j)) = \frac{a\cdot(o^k(i) - o^k(j))}{2}$ and $a = 1/(n-1)$, the transformation function TF_o^f can be represented as

$$TF_o^f(o^k(i), o^k(j)) = \frac{1}{2}\left(1 + \frac{o^k(j) - o^k(i)}{n-1}\right).$$

(ii) Transforming preference orderings into multiplicative preference relations.
Herrera et al. [21] proposed a transformation function between preference orderings and multiplicative preference relations.
For decision maker d_k, using transformation function TF_o^m to obtain his/her multiplicative preference value of alternative x_i over x_j, a_{ij}^k, from values $o^k(i)$ and $o^k(j)$, as Eq. (5):

$$a_{ij}^k = TF_o^m(o^k(i), o^k(j)) = 9^{[\varphi_4(n-o^k(i)) - \varphi_4(n-o^k(j))]}, \tag{5}$$

where φ_4 represents any decreasing and continuous function.
When setting $\varphi_4(n - o^k(i)) = \frac{n-o^k(i)}{n-1}$ [58], the transformation function TF_o^m can be represented as

$$TF_o^m(o^k(i), o^k(j)) = 9^{\frac{o^k(j)-o^k(i)}{n-1}}.$$

(3) **Transformation functions among fuzzy preference relations, multiplicative preference relations and linguistic preference relations**

The transformation functions among fuzzy preference relations, multiplicative preference relations and linguistic preference relations have been proposed in Chiclana et al. [12] and Dong et al. [20].

(i) Transformation functions between fuzzy preference relations and linguistic preference relations.
Let X, D and $\overline{S}$ be as before. Dong et al. [20] defined the additive scale function to complete the transformation between linguistic preference relations and fuzzy preference relations.

Definition 1 [20]. A monotonically increasing function δ: $\overline{S} \to [0,1]$ is called a additive scale function if $\delta(s) + \delta(neg(s)) = 1$, where $s \in \overline{S}$.

For example, the additive scale function of the linear scale is

$$\delta(s) = 0.5 + \frac{1}{g}(\Delta^{-1}(s) - g/2). \tag{6}$$

Let $L^k = \left[l_{ij}^k\right]_{n\times n}$ be a linguistic preference relation based on $\overline{S}$ and associated with decision maker d_k. Let δ be the additive scale function over $\overline{S}$, and let $P^k = \left[p_{ij}^k\right]_{n\times n}$ be a corresponding fuzzy preference relation associated with L^k. Then, the transformation function between P^k and L^k is as Eq. (7):

$$p_{ij}^k = \delta\left(l_{ij}^k\right). \tag{7}$$

Because the additive scale function δ is a monotonically increasing function, there is an inverse function of δ, denoted as δ^{-1}, which guarantees fuzzy preference relations can be transformed into linguistic preference relations.
Additionally, Herrera and Martínez [22] proposed a two-step method to transform [0, 1] values into a linguistic 2-tuple, which can be used to transform fuzzy preference relations into 2-tuple linguistic preference relations.

(ii) Transformation functions between multiplicative preference relations and linguistic preference relations.
Let X, D and $\overline{S}$ be as before. In [20], Dong et al. defined the multiplicative scale function to complete the transformation between multiplicative preference relations and linguistic preference relations.

Definition 2 [20]. A monotonically increasing function f: $\overline{S} \to R^+$ is called a multiplicative scale function if $f(s) \times f(neg(s)) = 1$, where $s \in \overline{S}$.

Note 1. Multiplicative scale defined over a pre-established linguistic label set, denoted by $\overline{S^{AHP}}$, are used in the analytic hierarchy process (AHP). There are several different numerical scales, such as the Saaty scale [13], the Ma-Zheng scale [30], the geometrical scale

[31] and the Salo-Hämäläinen scale [32], etc., to quantify $\overline{S^{AHP}}$ in AHP.

For example, the multiplicative scale function of the Saaty scale is

$$f(s) = \begin{cases} \Delta^{-1}(s) - 7, & s \geqslant s_8 \\ \frac{1}{9-\Delta^{-1}(s)}, & s < s_8 \end{cases}, \tag{8}$$

where $s \in \overline{S^{AHP}}$ and $\overline{S^{AHP}}$ is quantified by the Saaty scale.

Other familiar multiplicative numerical scales (i.e., the Ma-Zheng scale, the geometrical scale and the Salo-Hämäläinen scale) and their scale functions can be found in [20].

Let $L^k = \left[l_{ij}^k\right]_{n\times n}$ be a linguistic preference relation based on $\overline{S}$ and associated with decision maker d_k. Let f be the multiplicative scale function over $\overline{S}$, and let $A^k = \left[a_{ij}^k\right]_{n\times n}$ be a corresponding multiplicative preference relation associated with L^k. Then, the transformation function between A^k and L^k is as Eq. (9):

$$a_{ij}^k = f\left(l_{ij}^k\right). \tag{9}$$

Because the multiplicative scale function f is a monotonically increasing function, there is an inverse function of f, denoted as f^{-1}, which guarantees that multiplicative preference relations can be transformed into linguistic preference relations.

(iii) Transformation functions between fuzzy preference relations and multiplicative preference relations.

Based on Definitions 1 and 2, Dong et al. [20] proposed the transformation function between fuzzy preference relations and multiplicative preference relations as:

Let $A^k = \left[a_{ij}^k\right]_{n\times n}$ be a multiplicative preference relation using the multiplicative scale function f, and let $P^k = \left[p_{ij}^k\right]_{n\times n}$ be a fuzzy preference relation using the additive scale function δ. Then, the transformation function between A^k and P^k is as Eq. (10):

$$p_{ij}^k = \delta\left(f^{-1}\left(a_{ij}^k\right)\right). \tag{10}$$

For example, if the multiplicative scale is the Saaty scale and the additive scale is the linear scale, Eq. (10) will reduce to the transformation function proposed in Chiclana et al. [12]:

$$p_{ij}^k = \frac{1}{2}\left(1 + \log_9 a_{ij}^k\right). \tag{11}$$

Using the inverse operation of Eq. (10), fuzzy preference relations can be transformed into multiplicative preference relations.

3.3. Transformation functions among multigranular linguistic term sets

Generally, the first and most important issue in managing GDM problems defined in the multigranular linguistic context is unifying the multigranular linguistic preference information provided by different decision makers into a uniformed linguistic term set.

Two main approaches to unify the multigranular linguistic information have been proposed in Herrera and Martínez [18] and Herrera et al. [17], which we called the fusion approach based on a linguistic 2-tuple and the fusion approach based on membership functions, respectively.

(1) **The fusion approach based on a linguistic 2-tuple**

Using the 2-tuple linguistic representation model, Herrera and Martínez [18] proposed the transformation functions among multigranular linguistic term sets.

Defining $LH = \cup_t l(t, n(t))$ as a linguistic hierarchy, in which t is a number that indicates the level of the hierarchy and $n(t)$ is the granularity of the linguistic term set of the level t, let $S^{n(t)} = \left\{s_0^{n(t)}, \ldots, s_{n(t)-1}^{n(t)}\right\}$ be a linguistic term set that belongs to a level t of the hierarchy and has a granularity of uncertainty $n(t)$.

Let Δ and Δ^{-1} be as before, then the recursive transformation function between a linguistic label that belongs to level t and a label in level $t' = t + b$, with $b \in Z$, is defined as $TF_t^{t'} : l(t, n(t)) \to l(t', n(t'))$, where

$$TF_t^{t'}(s_i^{n(t)}, \alpha^{n(t)}) = \Delta\left(\frac{\Delta^{-1}(s_i^{n(t)}, \alpha^{n(t)}) \cdot (n(t') - 1)}{n(t) - 1}\right). \tag{12}$$

This fusion approach for multigranular linguistic information mentioned above has been used in [20,59]. Recently, Dong et al. [60] extended the 2-tuple linguistic preference relations into the multigranular unbalanced linguistic context proposed in [61].

(2) **The fusion approach based on membership functions**

Based on membership functions, Herrera et al. [17] proposed a fusion method for multigranular linguistic information.

Let $S_B = \left\{s_0^B, \ldots, s_p^B\right\}$ and $S_T = \left\{s_0^T, \ldots, s_q^T\right\}$ be any two linguistic term sets, where $p + 1$, $q + 1$ are the granularities of S_B, S_T, respectively. Let $F(S_T)$ be the fuzzy set in S_T. In [17], the transformation from S_B to $F(S_T)$ is defined as $TF_B^T : S_B \to F(S_T)$, where

$$TF_B^T(s_i^B) = \left\{\left(s_j^T, \gamma_j^i\right)/j \in \{0, \ldots, q\}\right\}, \forall s_i^B \in S_B, \tag{13}$$

$$\gamma_j^i = \max_y \min\left\{\mu_{s_i^B}(y), \mu_{s_j^T}(y)\right\},$$

where $\mu_{s_i^B}(y)$ and $\mu_{s_j^T}(y)$ are the membership functions of the fuzzy sets associated to the terms s_i^B and s_j^T, respectively. Obviously, the result of TF_B^T for any linguistic term of S_B is a fuzzy set defined in the S_T.

This fusion approach for multigranular linguistic information mentioned above has been widely used in [23,46,62–64].

It is easy to find that the membership function of each linguistic term is critical to the fusion approach mentioned above. In a linguistic hierarchy [65], no matter how many linguistic terms each level has, assuming every linguistic term in each level covers an interval $[a, b] \in [0, 1]$. Obviously, the interval that each linguistic term covers overlaps with the other intervals from other linguistic term sets. Therefore, the ratio of the common interval they cover indicates the membership function that should be assigned. Bearing this in mind, Chen and Ben-Arieh [66] proposed another fusion method for multigranular linguistic information.

To improve readability, Table 1 summarizes the transformation functions among the heterogeneous preference structures, which have been introduced in detail in Sections 3.2 and 3.3. In Table 1 the first column indicates one preference structure (i.e., preference structure I), the second column indicates another preference structure (i.e., preference structure II), and the third column shows the researches regarding the transformation functions between preference structures I and II.

Table 1
The transformation functions among the heterogeneous preference structures.

Preference structure I	Preference structure II	Transformation functions
Utility values	Preference orderings	Chiclana et al. [9] Büyüközkan et al. [67]
	Fuzzy preference relations	Luce and Suppes [68] Tanino [53,54] Dombi [69] Chiclana et al. [56,9] Chiclana et al. [12] Herrera-Viedma et al. [10] Pérez et al. [70]
	Multiplicative preference relations	Saaty [13,57] Herrera et al. [21] Zhang et al. [71]
Preference orderings	Fuzzy preference relations	Chiclana et al. [72] Chiclana et al. [56,9] Chiclana et al. [12] Herrera-Viedma et al. [10] Pérez et al. [70]
	Multiplicative preference relations	Herrera et al. [21] Zhang et al. [71]
Linguistic preference relations	Fuzzy preference relations	Delgado et al. [16] Büyüközkan et al. [67] Herrera and Martínez [22] Herrera et al. [23] Martínez et al. [73] Dong et al. [20]
	Multiplicative preference relations	Dong et al. [74] Dong et al. [20] Dong et al. [75]
Fuzzy preference relations	Multiplicative preference relations	Chiclana et al. [12] Herrera-Viedma et al. [10] Herrera-Viedma et al. [76] Wang and Fan [77] Wang and Parkan [78] Pérez et al. [70] Dong et al. [20]
Linguistic term sets with the granularity T_1	Linguistic term sets with the granularity T_2	Herrera et al. [17] Herrera-Viedma et al. [79] Herrera-Viedma et al. [46] Chang et al. [62] Wang [64] Mata et al. [63] Herrera and Martínez [18] Martínez et al. [73] Dong et al. [20] Porcel et al. [59] Dong et al. [60] Herrera et al. [23] Palomares et al. [47] Chen and Ben-Arieh [66] Xu [80]

4. The optimization-based approach

In this section, we review the optimization-based approach for GDM with heterogeneous preference structures. Specifically, Section 4.1 introduces the framework of the optimization-based approach, and surveys the optimization-based models presented in Ma et al. [26] and Jiang et al. [27]. Section 4.2 summaries the variations for the optimization-based approach.

4.1. Basic idea and model

In the optimization-based approach [24–27], multi-objective optimization models are constructed to integrate heterogeneous preference structures. The framework of the optimization-based approach to solve GDM problems with heterogeneous preference structures is described in Fig. 2.

Fan et al. [24,25], Ma et al. [26] and Jiang et al. [27] initiated the optimization-based approach to manage the GDM with heterogeneous preference structures. Next, we will introduce two optimization-based models proposed by Ma et al. [26] and Jiang et al. [27].

(1) **An optimization-based model for GDM with heterogeneous preference structures.**

Fan et al. [25] proposed an optimization model to handle GDM problems with two different preference structures (i.e., multiplicative preference relations and fuzzy preference relations), and Ma et al. [26] further incorporated utility values, preference orderings into the optimization model.

Let X and D be as before. In [26], Ma et al. assumed decision maker $d_k (k = 1, 2, \ldots, m)$ provides his/her opinion on X in one of the following preference structures: preference orderings $O^k = \{o^k(1), \ldots, o^k(n)\}$, utility values $U^k = \{u_1^k, u_2^k, \ldots, u_n^k\}$, fuzzy preference relation $P^k = \left(p_{ij}^k\right)_{n\times n}$ and multiplicative preference relation $A^k = \left(a_{ij}^k\right)_{n\times n}$.

Let D^U, D^O, D^A and D^F be four subsets of D, where $D^U \cup D^O \cup D^A \cup D^F = D$, to represent the decision makers whose preference

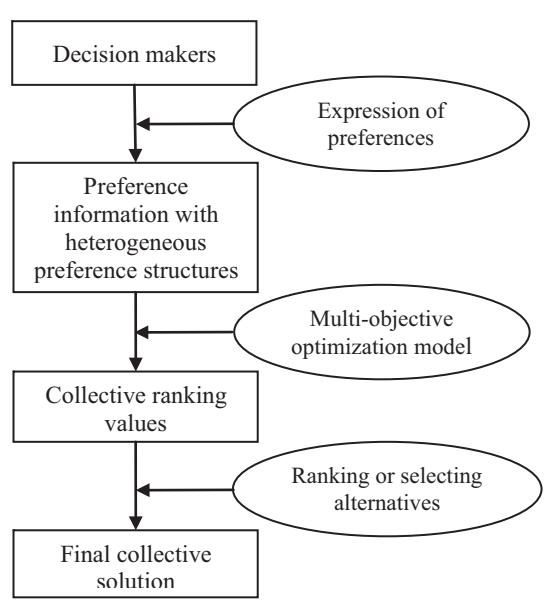

Fig. 2. The framework of the optimization-based approach.

structure on X is utility values, preference orderings, multiplicative preference relations and fuzzy preference relations, respectively. For simplicity, let $D^U = \{d_1, \ldots, d_{r_1}\}, D^O = \{d_{r_1+1}, \ldots, d_{r_2}\}$, $D^A = \{d_{r_2+1}, \ldots, d_{r_3}\}, D^F = \{d_{r_3+1}, \ldots, d_m\}$. The corresponding subscripts of these four subsets of D are denoted as $R^U = \{1, \ldots, r_1\}$, $R^O = \{r_1 + 1, \ldots, r_2\}$, $R^A = \{r_2 + 1, \ldots, r_3\}$ and $R^F = \{r_3 + 1, \ldots, m\}$, respectively.

Let w_i be the ranking value of the alternative $x_i \in X$ such that the larger the value of w_i, the more important the alternative x_i. To obtain $(w_1, w_2, \ldots, w_n)^T$, the basic idea of the optimization-based approach in [26] is shown as follows:

(i) For decision makers D^U, the ranking value of alternative x_i, w_i, is expected to be closed for the utility value given by each decision maker to alternative $x_i, u_i^r (i = 1, \ldots, n)$, where $r \in R^U$;
(ii) For decision makers D^O, the ranking value of alternative x_i, w_i, is expected to have perfect consistency for the preference ordering given by each decision maker to alternative x_i, $o^r(i)(i = 1, \ldots, n)$, where $r \in R^O$;
(iii) For decision makers D^A, the multiplicative preference relations a_{ij}^r are expected to have perfect consistency for the ratio of ranking values, $w_i/w_j \forall i, j \in \{1, \ldots, n\}$, $r \in R^A$;
(iv) For decision makers D^F, the fuzzy preference relations p_{ij}^r are expected to have perfect consistency for the ratio of ranking values, $w_i/(w_i + w_j) \forall i, j \in \{1, \ldots, n\}$, $r \in R^F$.

Let $C = (c_1, c_2, \ldots, c_m)^T$ be the weight vector, where c_k denotes the importance score of decision maker d_k, $c_k \geqslant 0$, and $\sum_{k=1}^m c_k = 1$. A four-objective constrained optimization model is then constructed to integrate four preference structures as model (14):

$$\begin{aligned}
&\min z_1 = \sum_{r \in R^U} c_r \sum_{i=1}^n \sum_{j=1}^n \left(w_i u_j^r - w_j u_i^r\right)^2 \\
&\min z_2 = \sum_{r \in R^O} c_r \sum_{i=1}^n \sum_{j=1}^n [w_i(n - o^r(j)) - w_j(n - o^r(i))]^2 \\
&\min z_3 = \sum_{r \in R^A} c_r \sum_{i=1}^n \sum_{j=1}^n [w_i - w_j a_{ij}^r]^2 \\
&\min z_4 = \sum_{r \in R^F} c_r \sum_{i=1}^n \sum_{j=1}^n \left[w_i - (w_i + w_j) p_{ij}^r\right]^2 \\
&s.t. \sum_{i=1}^n w_i = 1 \\
&w_i \geqslant 0, \quad i = 1, \ldots, n.
\end{aligned} \tag{14}$$

Based on the linear weighted summation method, this multi-objective optimization model can be converted to a single objective optimization model. The details of this optimization-based model can be found in [26]. By solving the optimization model, the final collective ranking values $(w_1, w_2, \ldots, w_n)^T$ on alternatives can be obtained.

(2) **An optimization-based model for GDM with multigranular linguistic information**

Jiang et al. [27] proposed an optimization-based approach to handle the GDM problem with multigranular linguistic information.

Let X and D be as before. Jiang et al. [27] assumed that different decision makers use multiple linguistic term sets with distinct granularities to express their individual preference information, i.e., according to his/her preferred linguistic term set

Table 2
Variations on the optimization-based approach.

References	Preference structures	Optimization models
Fan et al. [24]	Multiplicative preference relations Fuzzy preference relations	A linear goal programming model
Wang and Parkan [81]	Multiplicative preference relations Fuzzy preference relations	A linear goal programming model
Wang et al. [82]	Multiplicative preference relations Fuzzy preference relations	A chi-square optimization model
Xu [83]	(Interval) utility values (Interval) fuzzy preference relations (Interval) multiplicative preference relations	A linear goal programming model
Fan and Zhang [84]	Incomplete multiplicative preference relations Incomplete fuzzy preference relations Incomplete linguistic preference relations	A linear goal programming model
Xu [85]	Preference orderings Utility values (Incomplete) fuzzy preference relations (Incomplete) multiplicative preference relations	A nonlinear programming model
Xu [86]	Incomplete multiplicative preference relations Incomplete fuzzy preference relations Incomplete additive linguistic preference relations Incomplete multiplicative linguistic preference relations	A nonlinear optimization programming model
Xu [87]	Multiplicative preference relations Fuzzy preference relations	A nonlinear optimization model
Wan and Li [88]	Intuitionistic fuzzy sets Trapezoidal fuzzy numbers Intervals Real numbers	A fuzzy mathematical programming model
Zhang and Gao [89]	Uncertain multiplicative preference relations Uncertain fuzzy preference relations Uncertain linguistic preference relations Intuitionistic fuzzy preference relations	A linear bi-objective programming model

$S^k = \left\{s_0^k, s_1^k, \ldots, s_{T_k}^k\right\}$, the decision maker $d_k(k = 1, 2, \ldots, m)$ provides his/her opinions on X as a vector $V^k = (v_1^k, \ldots, v_n^k)$, where $v_i^k (i = 1, 2, \ldots, n) \in S^k$ denotes his/her opinion over the alternative x_i. The semantics of the term $v_i^k (i = 1, 2, \ldots, n)$ is described by a trapezoidal fuzzy number, i.e., $v_i^k = \left(v_i^{a,k}, v_i^{b,k}, v_i^{d,k}, v_i^{c,k}\right)$ $(i = 1, 2, \ldots, n)$.

Let $v_i (i = 1, 2, \ldots, n)$ be the collective ranking value of the alternatives x_i. The semantics of the term $v_i (i = 1, 2, \ldots, n)$ is described as $v_i = (v_i^a, v_i^b, v_i^d, v_i^c) (i = 1, 2, \ldots, n)$. To preserve decision makers' opinions as much as possible, an optimization model to minimize the deviation degree between each decision maker's opinion and the collective opinion is constructed as model (15):

$$\begin{cases} Min \;\; z = \frac{1}{4}\sum_{i=1}^{n}\sum_{k=1}^{m}\left(\left|v_i^{a,k} - v_i^a\right| + \left|v_i^{b,k} - v_i^b\right| + \left|v_i^{d,k} - v_i^d\right| + \left|v_i^{c,k} - v_i^c\right|\right) \\ s.t. \; 0 \leqslant v_i^a \leqslant v_i^b \leqslant v_i^d \leqslant v_i^c \leqslant 1, \;\; i = 1, \ldots, n. \end{cases} \tag{15}$$

This optimization model can be transformed into a linear goal optimization model. By solving the goal programming model (20), the final collective ranking values $V = (v_1, \ldots, v_n)^T$ on alternatives can be obtained. The details of this optimization-based model can be found in [27].

4.2. Variations for the optimization-based approach

Since Fan et al. [25], Ma et al. [26] and Jiang et al. [27] proposed the optimization-based approach for GDM with heterogeneous preference structures, many variations for the optimization-based approach have been presented over the last ten years. Table 2 summarizes these variations. The basic idea of these variations is similar to that proposed in [25–27], which has been introduced in detail in Section 4.1. The main difference among these variations lies in the use of different preference structures and different optimization objective functions. In Table 2 the second column indicates the heterogeneous preference structures discussed, and the third column shows the optimization models constructed.

5. Direct approach

In this section, we introduce the direct approach for GDM with heterogeneous preference structures. Specifically, Section 5.1 reviews a direct consensus framework. Section 5.2 investigates the application of the direct approach in AHP.

5.1. Direct consensus framework

Inspired by the direct approach initiated in Herrera et al. [28], the GDM model initiated in Chiclana et al. [9] and the corresponding consensus model presented in Herrera-Viedma et al. [10], Dong and Zhang [29] proposed a direct consensus framework to manage the GDM with heterogeneous preference structures, which is depicted in Fig. 3.

There are some differences between the direct approach proposed in [29] and the indirect approach presented in [9,10]. Specifically, the differences manifest in three aspects:

(1) The selection process.
The selection process in the framework of the indirect approach has been introduced in Section 3. Unlike the selection process in the indirect approach [9], the implementation of the selection process in the direct approach [29] comprises two steps: (1) obtaining individual preference vectors by using the corresponding individual selection methods; and (2) obtaining a collective preference vector by aggregating all the standardized individual preference vectors.
Thus far, many methods have been proposed to derive the individual preference vector from preference relations (e.g., [13,14,90–99]).

(2) The consensus process.
Compared with the consensus process presented in the indirect approach [10], Dong and Zhang [29] proposed novel feedback adjustment rules associated with individual preference representation structures that consist of two steps: (1) the collective preference vector is transformed into the preference information that is described by the preference representation structure used by the individual decision maker; and (2) decision makers revised his/her preferences, according to the original individual preference information and the transformed collective preference information.

(3) Internal consistency and Pareto principle.

Internal consistency means that the rankings among alternatives derived from the transformed preference representation structure is the same one from the original preference representation structure. Dong and Zhang [29] show that the selection process proposed in the indirect approach [9,10] may not satisfy the internal consistency and the famous Pareto principle of social choice theory. The direct consensus framework presented in [29] can avoid the internal inconsistency issue and satisfy the Pareto principle of social choice theory.

5.2. Application of the direct approach in AHP

Since the appearance of AHP, the AHP-GDM problem has been a popular topic [100–103]. In traditional AHP-GDM problems, only one numerical scale and one prioritization method are employed; however, the AHP linguistic scale that is used to construct a linguistic pairwise comparison matrix (LPCM), i.e., linguistic preference relations, may have different numerical meanings to different individuals [104,105]. Therefore, a new hypothesis is naturally proposed for AHP-GDM problems: different numerical scales and prioritization methods are needed for different LPCMs. Based on this hypothesis, there are two challenges in AHP-GDM:

(1) How to design a procedure to automatically set different individual numerical scales and prioritization methods for different LPCMs in AHP.
(2) How to assist AHP decision makers, who use different numerical scales and prioritization methods, to rank the alternatives or reach a consensus.

Approaches to handling the first challenge have been proposed in [75,106–108]. In particular, Dong et al. [106] proposed a 2-tuple linguistic based multicriteria approach to set the individual numerical scale and prioritization method for different LPCMs, in which two basic concepts of the linguistic Euclidean distance (LED) and linguistic minimum violations (LMV) are used.

Let S and Δ^{-1} be as before, and $g + 1$ is the granularity of S. Suppose $C = (c_{ij})_{n\times n}$ and $E = (e_{ij})_{n\times n}$ are any two LPCMs, where $c_{ij}, e_{ij} \in S$. Then

$$(1)\; LED(C, E) = \frac{1}{n}\sqrt{\sum_{j=i}^{n}\sum_{i=1}^{n}\left(\frac{\Delta^{-1}(c_{ij}) - \Delta^{-1}(e_{ij})}{g+1}\right)^2}, \tag{16}$$

$$(2)\; LMV(C, E) = \frac{1}{n}\sum_{j=i}^{n}\sum_{i=1}^{n} I_{ij}, \tag{17}$$

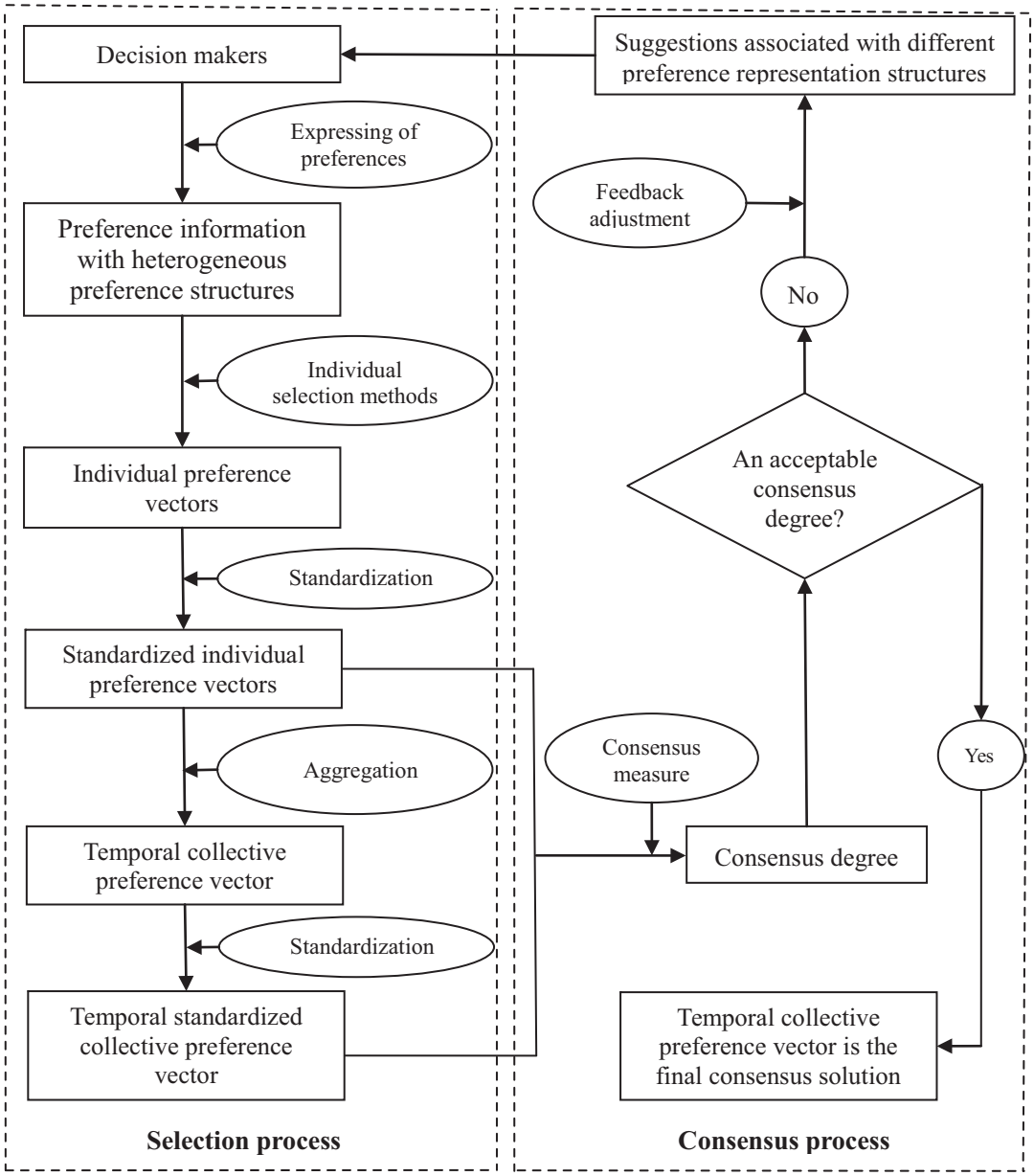

Fig. 3. The direct consensus framework of GDM with heterogeneous preference structures.

where

$$I_{ij} = \begin{cases} 1 & \text{if } c_{ij} > s_{g/2} \quad \text{and } e_{ij} < s_{g/2} \\ 1 & \text{if } c_{ij} < s_{g/2} \quad \text{and } e_{ij} > s_{g/2} \\ 0.5 & \text{if } c_{ij} = s_{g/2} \quad \text{and } e_{ij} \neq s_{g/2} \\ 0.5 & \text{if } c_{ij} \neq s_{g/2} \quad \text{and } e_{ij} = s_{g/2} \\ 0 & \text{otherwise} \end{cases}.$$

Smaller values of the LED and LMV indicate smaller differences between two LPCMs.

Let X, D and $\overline{S^{AHP}}$ be as before, and let $L^k = \left[l_{ij}^k\right]_{n\times n}$ be a LPCM based on $\overline{S^{AHP}}$, associated with decision maker d_k. Let $NS = \{ns^1, \ldots, ns^i, \ldots, ns^I\}$, where ns^i represents the ith candidate numerical scale, and $PM = \{pm^1, \ldots, pm^j, \ldots, pm^J\}$, where pm^j represents the jth candidate prioritization method. Then, the 2-tuples linguistic based multicriteria approach proposed in Dong et al. [106] can be described as follows:

Input: The LPCM L^k, the set NS, and the set PM.
Output: The matrix $TP = (tp_{ij})_{I\times J}$, the individual numerical scale of d_k, i.e., f^k, and the individual prioritization method of d_k, i.e., o^k.
Step 1: Apply each numerical scale $ns^p \in NS$ and each prioritization method $pm^q \in PM$ to the LPCM L^k, and obtain the corresponding individual priority vector for X,

$$w^{(k,p,q)} = \left(w_1^{(k,p,q)}, w_2^{(k,p,q)}, \ldots, w_n^{(k,p,q)}\right)^T.$$

Step 2: Based on the inverse operation of the selected numerical scale function (see [106]), the LPCM of $w^{(k,\ p,\ q)}$, i.e., $\overline{L^{(k*,p,q)}}$, can be obtained.
Step 3: Let $A = (a_{pq})_{I\times J}$, where $a_{pq} = LED(L^k, \overline{L^{(k*,p,q)}})$. Let $B = (b_{pq})_{I\times J}$, where $b_{pq} = LWV(L^k, \overline{L^{(k*,p,q)}})$.
Step 4: Normalize the matrices A and B to obtain the normalized matrices $\overline{A} = (\overline{a_{ij}})_{I\times J}$ and $\overline{B} = (\overline{b_{ij}})_{I\times J}$ as follows:

$$\overline{a_{ij}} = \frac{\max_{i,j} a_{ij} - a_{ij}}{\max_{i,j} a_{ij} - \min_{i,j} a_{ij}},\ i = 1,2,\ldots,I,\ j = 1,2,\ldots,J,$$

$$\overline{b_{ij}} = \frac{\max_{i,j} b_{ij} - b_{ij}}{\max_{i,j} b_{ij} - \min_{i,j} b_{ij}},\ i = 1,2,\ldots,I,\ j = 1,2,\ldots,J.$$

Step 5: Let $TP = (tp_{ij})_{I\times J}$, where $tp_{ij} = \frac{1}{2}\overline{a_{ij}} + \frac{1}{2}\overline{b_{ij}}$.

Step 6: Output the matrix TP. If $tp_{s\tau} = \max_{i,\,j} tp_{ij}$, then the individual numerical scale $f^k = ns^s$ and the individual prioritization method $o^k = pm^\tau$.

Based on the 2-tuple linguistic based multicriteria approach presented in [106], Dong et al. [109] handled the second challenge mentioned above by exploring an application of the direct approach in AHP; in this way, a novel consensus model in a local context is proposed to manage the AHP-GDM with different individual numerical scales and prioritization methods.

6. Summary, critical discussions and new directions

The management of heterogeneous preference structures is always hard and complex in real-world decision problems. Three types of fusion approaches (i.e., indirect approach, optimization-based approach, and direct approach) have been proposed to manage heterogeneous preference information.

In the indirect approach, transformation functions are used to unify the heterogeneous preference structures. The rankings of alternatives are derived from the collective preference information obtained by aggregating uniformed individual preference information.

In the optimization-based approach, multi-objective optimization models are constructed to integrate the individual preference information with heterogeneous preference structures. Solving the optimization models yields the rankings of alternatives.

In the direct approach, the individual preference vectors are derived by individual selection methods, and aggregating individual preference vectors obtains the rankings of alternatives.

From the perspective of internal consistency and the Pareto principle of social choice theory, some results regarding the three types of fusion approaches have been proposed in [12,19,20,29], which are summarized in Table 3.

Although the three types of fusion approaches have been introduced in the specialized literatures to manage the heterogeneous preference structures in GDM, there are some weaknesses that must be highlighted.

(1) Although some comparison results (see Table 3) have been proposed for the three types of fusion approaches, different fusion approaches may have different advantages and drawbacks in different decision-making scenarios. Existing studies do not provide sufficient comparisons among the three types of fusion approaches.
(2) Consensus reaching processes have been investigated in the indirect approach and the direct approach in depth, which guarantee a high consensus degree among the decision makers. However, the solutions to the optimization models in the optimization-based approach cannot guarantee a high consensus degree among the decision makers.
(3) The GDM with heterogeneous preference structures is based on a natural premise: the decision makers who have different experiential, cultural and educational backgrounds will provide their preference information by heterogeneous preference structures. In contrast to the large number of theoretical studies, however, there are relatively few works regarding the applications of the three types of fusion approaches (see [110–112]).
(4) Decision making not only relates to mathematical models, but also to philosophical issues; however, philosophical foundations have never been considered in the GDM with heterogeneous preference structures.

Table 3
The comparison among the three types of fusion approaches.

Fusion approaches	Avoids the internal inconsistency?	Satisfies the Pareto principle?
Indirect approach	Need to satisfy some conditions	No
Optimization-based approach	Yes	Unknown
Direct approach	Yes	Yes

Thus, the following directions should be considered for further research:

(1) It is necessary to justify the comparative criteria, and propose a detailed comparative study based on established criteria to adequately evaluate the advantages and drawbacks of the three types of fusion approaches.
(2) It would be useful to carry out a theoretical development of the existing transformation functions to justify their advantages to unify heterogeneous preference information.
(3) It would be interesting to integrate the consensus process into the optimization-based approach in future research.
(4) As a new trend, it is the application of the theoretical models to real-world GDM problems with heterogeneous preference structures.
(5) The use of philosophical foundations to improve the fusion process of GDM with heterogeneous preference structures should be investigated in future research.

7. Conclusions

This paper reviews and analyzes three types of fusion approaches (i.e., the indirect approach, the optimization-based approach and the direct approach) to manage the fusion process in GDM with heterogeneous preference structures. Specifically, we review the different transformation functions among utility values, preference orderings, numerical preference relations, and multigranular linguistic preference relations. Next, we review the origination of the optimization-based approach and summarize the variations for the optimization-based models, which are based on the use of different preference structures and different optimization objective functions. Subsequently, we review the direct consensus framework and the application of the direct approach in AHP. Finally, we summarize the characteristics of the three types of fusion approaches and note several critical comments and new directions that are ripe for future research.

Acknowledgements

Yucheng Dong would like to acknowledge the financial support of a Grant (No. 71171160) from NSF of China, and a Grant (No. skqx201308) from Sichuan University.

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